The documentation and process conversion measures necessary to comply with this document shall be completed by 21 October 2021.

INCH-POUND

MIL-PRF-19500/383C
21 July 2021
SUPERSEDING
MIL-PRF-19500/383B
w/AMENDMENT 1
15 April 2000

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE. SILICON, VOLTAGE-VARIABLE CAPACITOR
TYPES 1N5131A THROUGH 1N5148A
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, voltage-variable-capacitor diode. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

* 1.2 Package outlines. See figure 1 (TO-7).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Limit	P_T 1/	V_{RM} (wkg)	V_{BR} I_R = 10 A dc	C	Capacitance Ratio	Q F = 50 MHz
	MW	V(pk)	V dc	pF		
Minimum	---	---	65	(See table II)	(See table II)	(See table II)
Maximum	400	60	---	(See table II)		

1/ Derate linearly 2.67 mW/°C above 25°C.

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AMSC N/A

FSC 5961

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* 1.4 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

* 1.4.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, and "JANTXV".

* 1.4.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

* 1.4.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.4.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "5139" through "5148".

* 1.4.2.3 Suffix symbols. The suffix letter "A" indicates a modified version of the non-suffix device type.

* 1.4.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

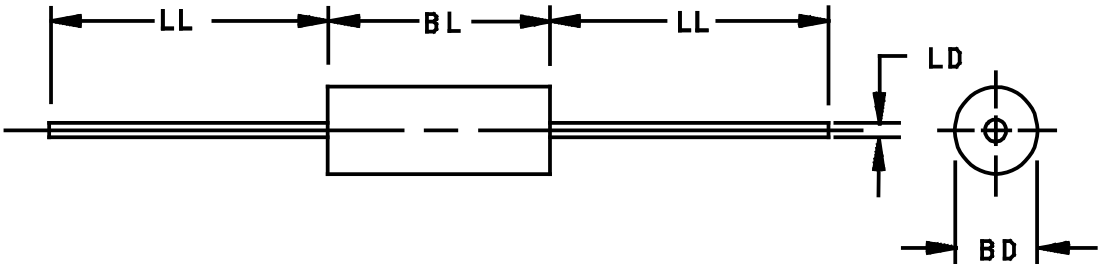
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://quicksearch.dla.mil.

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	0.230	0.300	5.84	7.62	
BD	0.085	0.107	2.16	2.72	5
LL	1.00	1.500	25.40	38.10	
LD	0.018	0.022	0.46	0.56	3, 4

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3 The specified lead diameter applies in the zone between 0.050 (1.27 mm) and 1.000 (25.40 mm) from the diode body. Outside of this zone the lead diameter is not controlled.
4 Both leads shall be within the specified dimensions.
5 The minimum body diameter shall be maintained over 0.15 (0.38 mm) inch of body length.
* 6. In accordance with ASME Y14.5, diameters are equivalent to Φx symbology.

FIGURE 1. Dimensions and configuration (TO-7).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.4.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

4. VERIFICATION

* 4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4, and tables I and II).

4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Lot accumulation period shall be 6 months in lieu of 6 weeks.

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Qualification for a particular group of capacitances. Qualification for a particular group of capacitances requires the testing of the lowest- and highest-nominal-capacitance units in the group.

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* 4.3 Screening (list applicable JAN levels). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements
	JANTX, JANTXV levels
2	T_A = +200°C, 48 hours
10	Not applicable
11	$I_{R;}$
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_R = 100 percent of initial value or 5 nA dc whichever is greater.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = 175°C; V_R = 60 V dc, 96 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition C, T (high) = +175°C.
B2	1056	Test condition A.
B3	1027	V_R = 60 V dc, T_A = 150°C, t = 500 hours.
B6	1032	T_A = 200°C, t = 500 hours.
B7		Temperature coefficient of capacitance. See 4.5.2 herein. T_A = -40°C to +85°C, max 0.03 % / °C. V_R = 4 V dc.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition A, 4 pounds, t = 15.
C2	2036	Test condition E.
C3	2016	1500 G, 0.5 ms, 5 blows in each orientation: X_1, Y_1, and Y_2.
C3	2006	20,000 G, in each orientation: X_1, Y_1, and Y_2.
C6	1026	V_R = 60 V dc, T_A = 150°C

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Temperature coefficient of capacitance test. Throughout the temperature range specified, the capacitance shall not change by more than the amount specified relative to the capacitance value measured at T_A = 25°C. The temperature coefficient of capacitance may be computed by the following formula:

$$TC_C = \frac{CT(+85°C) - (-40°C)}{85 + 40} \times \frac{100}{CT(25°C)}$$

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage	4021	I_R = 10 μA dc	B_V	65		V dc
Reverse current	4016	V_R = 55 V dc; dc method	I_R		20	nA dc
Subgroup 3						
Reverse current	4016	V_R = 55 V dc; dc method; T_A = 150°C	I_R		20	μA dc
Subgroup 4						
Capacitance ratio operation	4001	From V_R = 4 V dc to V_R = 60 V dc; f = 1 MHz		Column 5		
Capacitance 3/	4001	V_R = 4 V dc; f = 1 MHz	C	Column 3	Column 4	pF
Quality factor	4036	V_R = 4 V dc; f = 50 MHz	Q	Column 6		

1/ For sampling plans, see MIL-PRF-19500.
2/ Column references are to table II herein.
3/ End point only.

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TABLE II. Test ratings.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
Type (See 1.1)	Capacitance V_R = 4 V dc			Capacitance ratio From V_R = 4 V dc to V_R = 60 V dc	Q V_R = 4 V dc f = 50 MHz
	Nom	Min	Max	Min	Min
	pF	pF	pF		
1N5139A	6.8	6.46	7.14	2.7	350
1N5140A	10.0	9.50	10.50	2.8	300
1N5141A	12.0	11.40	12.60	2.8	300
1N5142A	15.0	14.30	15.70	2.8	250
1N5143A	18.0	17.10	18.90	2.8	250
1N5144A	22.0	20.90	23.10	3.2	200
1N5145A	27.0	25.70	28.30	3.2	200
1N5146A	33.0	31.40	34.60	3.2	200
1N5147A	39.0	37.10	40.90	3.2	200
1N5148A	47.0	44.70	49.30	3.2	200

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

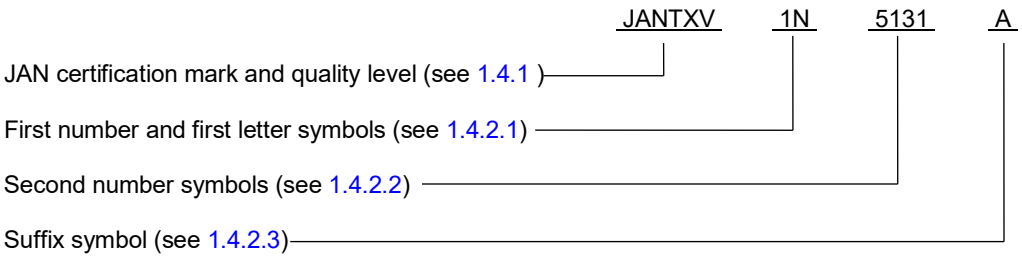
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.4 and 6.5.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: /VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

* 6.4 PIN construction example. The PINs are constructed using the following form.



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* 6.5 List of PINs.

Quality level JAN		Quality level JANTX		Quality level JANTXV	
JAN1N5131A	JAN1N5141A	JANTX1N5131A	JANTX1N5141A	JANTXV1N5131A	JANTXV1N5141A
JAN1N5132A	JAN1N5142A	JANTX1N5132A	JANTX1N5142A	JANTXV1N5132A	JATXVN1N5142A
JAN1N5134A	JAN1N5143A	JANTX1N5134A	JANTX1N5143A	JANTXV1N5134A	JANTXV1N5143A
JAN1N5135A	JAN1N5144A	JANTX1N5135A	JANTX1N5144A	JANTXV1N5135A	JANTXV1N5144A
JAN1N5136A	JAN1N5145A	JAN1N5136A	JANTX1N5145A	JANTXV1N5136A	JANTXV1N5145A
JAN1N5137A	JAN1N5146A	JANTX1N5137A	JANTX1N5146A	JANTXV1N5137A	JANTXV1N5146A
JAN1N5138A	JAN1N5147A	JANTX1N5138A	JANTX1N5147A	JANTXV1N5138A	JANTXV1N5147A
JAN1N5139A	JAN1N5148A	JANTX1N5139A	JANTX1N5148A	JANTXV1N5139A	JANTXV1N5148A
JAN1N5140A		JANTX1N5140A		JANTXV1N5140A	

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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* Air Force - 85
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